WEB DEVELOPMENT STANDARD

Develop HTML Frontend Interface

Mba'éichapa Ojejapo HTML Frontend Ne'ẽmbyrenda Iporãva, Isa'ykatúva ha Ipyahúva

Kuatia Rehegua: Kuatiañe'ẽ HTML Frontend Rekorã

Mata Duree: HTML5, Responsive Layout, CSS Grid, Form Handling

Avakuérape Guarã: Web Developers, Frontend Engineers, UI/UX Designers

Tekotevẽ: Mborayhu ha Tekopyahu / Approved Standard

1. Mba'épa HTML Frontend Interface Development

HTML (HyperText Markup Language) ha'e pe ypytyvõ guasu oñemombaretéva reheve opaichagua ñanduti rendague (website). Frontend interface ijapopyre ha'e pe tembiporu ha ta'anga puruhára (user) ohecha ha ojapóva peteĩ apopyre internet-pe.

1.1 Tekotevẽva HTML5-pe

HTML5 ogueru heta mba'e pyahu ikatúva ojeporu ombohapévo tekorã iporãva ha isãmbyhýva katuete:

Mba'e guasu HTML5 ijpyahúva:

- **Semantic Elements:** Tag-kuéra oikuaaukáva hekopete pe contenido (`<header>`, `<nav>`, `<section>`, `<article>`, `<footer>`).
- **Media Support:** Mbarete omba'apo haguã ta'angamýi ha purory ndive (`<video>`, `<audio>`).
- **Form Validation:** Ojehecha tekotevẽva oñemboty haguã formulariokuéra hekopete.

2. HTML Structural Architecture

HTML5 apopyre ojeporu katuete oñembohapa haguã pe kuatiarenda (webpage). Apytu'ũ ha rete oñemohenda porã va'erã.

2.1 Visual Example Matrix

```
<!DOCTYPE html>
<html lang="gn">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Frontend HTML Interface</title>
</head>
<body>
    <header>
        <h1>Frontend App Interface</h1>
    </header>
    <main>
        <section>
            <h2>Mba'e pyahu</h2>
            <p>Kóva ha'e peteĩ frontend rehegua.</p>
        </section>
    </main>
    <footer>
        <p>© 2026 Frontend Dev</p>
    </footer>
</body>
</html>
```

3. HTML Form Design & User Inputs

Formulariokuéra ha'e pe mba'e tenondegua oñekomunika haguã puruhára ha backend ndive.

3.1 Login ha Registration Forms

```
<form action="/api/login" method="POST" class="form-card">
    <div class="form-group">
        <label for="email">I-mail:</label>
        <input type="email" id="email" name="email" required placeholder="tech@example.com">
    </div>
    <div class="form-group">
        <label for="password">Ñe'ẽñemi:</label>
        <input type="password" id="password" name="password" required minlength="8">
    </div>
    <button type="submit" class="btn-primary">Eike</button>
</form>
```

4. CSS Styling & Layout Mechanisms

HTML rendaguépe ojeporu CSS (Cascading Style Sheets) oñembosa'y ha oñemohenda haguã ta'angakuéra.

4.1 Flexbox Layout

```
.navbar {
    display: flex;
    justify-content: space-between;
    align-items: center;
    background-color: #0f172a;
    padding: 1rem 2rem;
    color: #ffffff;
}

.nav-links {
    display: flex;
    gap: 1.5rem;
    list-style: none;
}
```

5. CSS Grid Architecture

Dashboard ha panel-kuéra kompleho omba'apo porãve CSS Grid oipurúvo.

```
.dashboard-grid {
    display: grid;
    grid-template-columns: 250px 1fr;
    grid-template-rows: auto 1fr auto;
    min-height: 100vh;
}

.sidebar {
    background-color: #1e293b;
    color: white;
}
```

6. Responsive Web Design (RWD)

Kuatiarendakuéra ojehecha va'erã porã celular, tablet ha computadora-pe.

```
@media (max-width: 768px) {
    .dashboard-grid {
        grid-template-columns: 1fr;
    }
    .sidebar {
        display: none;
    }
}
```

7. Web Accessibility (a11y) & ARIA

Opaite ava oguereko va'erã tembiporu oiporu haguã pe frontend interface.

Accessibility Rules:

- Ojeporu va'erã `alt` tag ta'angakuérape.
- Form control-kuéra oguereko va'erã `<label>` hekopete.
- Color contrast ojehecha va'erã oñemoñe'ẽ porã haguã.

8. JavaScript DOM Interaction

JavaScript ombovy'a ha omboyke kuatiarenda HTML.

```
document.querySelector('form').addEventListener('submit', (e) => {
    e.preventDefault();
    console.log('Formulario oñembohatái!');
});
```

9. Mohu'ã ha Tekorã Frontend-pe

HTML Frontend Interface apo ojerure arandu HTML5, CSS layout, accessibility ha responsiveness rehegua.

Mba'e	Tekorã Teñói	Tekorã Pyahu
Layout	HTML Tables	Flexbox & CSS Grid
Semantics	Div-heavy Markup	HTML5 Semantic Tags